KUET · ESE 4115 · Power Plant Engineering

ESE 4115 Question Map

Every question from the 2020, 2021, 2022, 2023, 2024 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 4115 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 6 h for Part A and 6 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×5, 2023 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: the 2024 exergy numericals (B7, 25 marks each) appeared only once, so they sit low. They are long; learn them only after the pass line. The Brayton calculator (B5) and the hydro calculator (A14) let you redo every year's numbers in minutes, so practise those first: A3, A8 and A14 are numericals that come almost every year.

Section A: Fundamentals, load curves, economics, forecasting, Bangladesh, hydro

Block 1: very likely and easy · 2.5 h

Topic	Chance	Level	Covered
A10 Load forecasting: need, factors, methods (similar-day, additive, peak vs energy)	Very likely 100%	Easy	11%
A4 Effects of variable load on power plant design and operation	Very likely 100%	Easy	18%

A2 Load curve, load duration curve and the plant factors (load, demand, diversity, capacity, use)	Very likely 93%	Easy	25%
A9 Tariff: objectives, requirements and types (flat, block, two-part, maximum demand, ToU)	Very likely 93%	Easy	31%
A7 Cost of generation, fixed and operating costs, depreciation	Very likely 87%	Easy	36%
A5 Site selection for a power plant and the interconnected grid	Very likely 80%	Easy	43%

Block 2: very likely, medium effort · 3.8 h

Topic	Chance	Level	Covered
A12 Hydro-electric plant: layout, essential features, dam, spillway, surge tank, forebay, hydrology terms, classification	Very likely 100%	Medium	54%
A14 Numericals: hydro power from catchment, rainfall and head	Very likely 100%	Medium	63%
A3 Numericals on load curve, load factor, capacity factor, use factor	Very likely 100%	Medium	70%
A13 Hydraulic turbines: selection, specific speed, governing of impulse (Pelton) turbine	Very likely 100%	Medium	77%
✔ Pass line: 77% of Section A marks covered, enough to choose 3 of the 4 questions (about 5.5 h)			
A11 Bangladesh power sector: fuel mix, generation cost, renewables, PSMP 2015, nuclear and coal plans	Very likely 93%	Medium	86%

Block 4: likely · 0.4 h

Topic	Chance	Level	Covered
A1 Power plant basics: classification, energy sources, base vs peak plant, choosing capacity	Likely 67%	Easy	93%

Only if time is left · 1.8 h

Topic	Chance	Level	Covered
A8 Numericals: generating cost per kWh	Possible 47%	Medium	99%
A6 Economic load sharing between a base load and a peak load plant (derivation)	Possible 33%	Hard	100%

Section B: Diesel, gas turbine, combined cycle, steam plant and auxiliaries

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered

B1 Diesel power plant: layout, working, fuel system, applications, advantages, peak-load use, lubrication	Very likely 100%	Easy	15%
B10 Feed water: impurities, troubles in boiler, treatment methods (lime-soda, zeolite)	Very likely 93%	Easy	23%
B6 Combined cycle power plant (gas turbine + steam turbine)	Very likely 93%	Easy	31%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B4 Improving gas turbine efficiency: regeneration, intercooling, reheating (and all three together)	Very likely 100%	Medium	40%
B11 Draught: natural vs mechanical (forced, induced, balanced), chimney and its height formula	Very likely 100%	Medium	46%
B13 Cooling towers: types, induced-draught counterflow wet tower, range, approach, efficiency + numerical	Very likely 93%	Medium	53%

Block 4: likely · 2.2 h

Topic	Chance	Level	Covered
B3 Gas turbine plant: open-cycle layout, open vs closed, applications, peak-load advantages, fuel properties	Likely 60%	Easy	57%
B8 Steam power plant: layout and four circuits, working, base-load advantages, site, design activities	Likely 67%	Medium	69%
B5 Brayton cycle numericals and the maximum-work condition	Likely 67%	Hard	76%

✔ Pass line: 76% of Section B marks covered, enough to choose 3 of the 4 questions (about 5.7 h)

Only if time is left · 2.9 h

Topic	Chance	Level	Covered
B2 Diesel plant cooling (necessity, thermostat cooling) and starting methods	Possible 53%	Easy	84%
B9 Reheat Rankine cycle and binary vapour cycle	Possible 53%	Medium	90%
B12 Numericals: chimney draught and chimney height	Possible 33%	Medium	92%
B7 Exergy analysis numericals (2024): Diesel cycle and closed Brayton cycle	Possible 33%	Hard	100%

Pass plan

2021-2024 papers: 4 questions of 35 marks per section, answer 3 per section. 2020 was a short paper (3 questions of 30 per section, answer 2). Section A = fundamentals, load curves and factors, economics and tariff, load forecasting, Bangladesh power sector, hydro-electric plant. Section B = diesel, gas turbine, combined cycle, steam plant, feed water, draught/chimney and cooling tower. Materials are Prof. Dr. Md. Shahidul Islam's slides (lectures 1-3, Bangladesh energy, hydro, load forecasting, diesel, gas turbine, steam plant, GT problem sheet). "Must study" topics came up in 3 or more of the 5 papers. Every numerical has a step table: left column is what you write, right column says why, in plain words.

SECTION A: FUNDAMENTALS, LOAD CURVES, ECONOMICS, FORECASTING, BANGLADESH, HYDRO

13 topics

cover 99% of all Section A marks asked

- Definitions, load curve, factors + factor numerical
- Variable load, site selection, grid; costs, depreciation, tariff + cost numerical
- Load forecasting and Bangladesh power sector
- Hydro plant, dams, turbines, governing + hydro numerical

SECTION B: DIESEL, GAS TURBINE, COMBINED CYCLE, STEAM PLANT AND AUXILIARIES

9 topics

cover 81% of all Section B marks asked

- Diesel plant layout, fuel, cooling, starting
- Gas turbine: layout, efficiency improvement, Brayton numerical
- Combined cycle, steam plant layout, reheat and binary cycles
- Feed water treatment, draught/chimney, cooling tower

- **Section A is almost free marks:** factors and load curves (A2, A3), effects of variable load (A4, asked every year), tariff (A9), load forecasting (A10) and Bangladesh PSMP (A11) repeat with the same wording.
- **Learn three numerical recipes for Section A:** plant factors from a load curve (A3), cost per kWh (A8) and hydro power $P = 9.81\ Q\ H\ \eta$ (A14). Each is 4-6 lines and comes every year.
- **Section B:** diesel plant layout + cooling (B1, B2), GT efficiency improvement (B4), combined cycle (B6), steam plant layout (B8), feed water (B10), draught (B11) and cooling tower (B13) cover nearly every question.

- **Brayton numerical (B5)** uses only two formulas: $T2/T1 = rp^{0.286}$ and $w = cp\ \Delta T$. The 2024 exergy problems (B7) are long but follow one fixed table: states, then work/heat, then entropy generation, then exergy.
- **Always draw:** load curve + LDC, hydro plant layout, diesel plant layout, thermostat cooling, open-cycle GT, T-s of Brayton with regeneration, combined cycle, steam plant four circuits, balanced draught, induced-draught cooling tower.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: FUNDAMENTALS, LOAD CURVES, ECONOMICS, FORECASTING, BANGLADESI PER PAPER

Topic	2024	2023	2022	2021	2020	Years	Marks	
A10 Load forecasting: need, factors, methods (similar-day, additive, peak vs energy)	16	18	18	16	11	5/5	79	Must st
A4 Effects of variable load on power plant design and operation	10	9	9	9	10	5/5	47	Must st
A2 Load curve, load duration curve and the plant factors (load, demand, diversity, capacity, use)	15	6	12	17		4/5	50	Must s
A9 Tariff: objectives, requirements and types (flat, block, two-part, maximum demand, ToU)	9	8	8	12		4/5	37	Must s
A7 Cost of generation, fixed and operating costs, depreciation	10	8	12		8	4/5	38	Must s
A5 Site selection for a power plant and the interconnected grid	10	10		18	8	4/5	46	Must s
A12 Hydro-electric plant: layout, essential features, dam, spillway,	17	17	19	14	6	5/5	73	Must stu

surge tank, forebay, hydrology terms, classification								
A14 Numericals: hydro power from catchment, rainfall and head	10	11	11	15	15	5/5	62	Must stu
A3 Numericals on load curve, load factor, capacity factor, use factor	15	9	11	12	8	5/5	55	Must stu
A13 Hydraulic turbines: selection, specific speed, governing of impulse (Pelton) turbine	8	7	7	7	14	5/5	43	Must stu
A11 Bangladesh power sector: fuel mix, generation cost, renewables, PSMP 2015, nuclear and coal plans	10	20	21	11		4/5	62	Must stu
A1 Power plant basics: classification, energy sources, base vs peak plant, choosing capacity		15	18	10	10	4/5	53	Mus
A8 Numericals: generating cost per kWh		11		12	13	3/5	36	Must st
A6 Economic load sharing between a base load and a peak load plant (derivation)	10					1/5	10	Skip if

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: DIESEL, GAS TURBINE, COMBINED CYCLE, STEAM PLANT AND AUXILIARIES:

Topic	2024	2023	2022	2021	2020	Years	Marks	
B1 Diesel power plant: layout, working, fuel system, applications, advantages, peak-load use, lubrication	10	22	20	35	10	5/5	97	Must st
B10 Feed water: impurities, troubles in boiler, treatment methods (lime-soda, zeolite)	18	11	12	15		4/5	56	Must s
B6 Combined cycle power plant (gas turbine + steam turbine)	10	10	14	15		4/5	49	Must s

B4 Improving gas turbine efficiency: regeneration, intercooling, reheating (and all three together)	10	12	16	15	12	5/5	65	Must stu
B11 Draught: natural vs mechanical (forced, induced, balanced), chimney and its height formula	5	8	8	10	8	5/5	39	Must stu
B13 Cooling towers: types, induced-draught counterflow wet tower, range, approach, efficiency + numerical	12	9	15	10		4/5	46	Must stu
B3 Gas turbine plant: open-cycle layout, open vs closed, applications, peak-load advantages, fuel properties		15	5	5		3/5	25	Good
B8 Steam power plant: layout and four circuits, working, base-load advantages, site, design activities		23	15	25	13	4/5	76	Must s
B5 Brayton cycle numericals and the maximum-work condition		10	12	15	14	4/5	51	Mus
B2 Diesel plant cooling (necessity, thermostat cooling) and starting methods		18	15		20	3/5	53	Must
B9 Reheat Rankine cycle and binary vapour cycle	25		11			2/5	36	Good to
B12 Numericals: chimney draught and chimney height		8			9	2/5	17	Good to
B7 Exergy analysis numericals (2024): Diesel cycle and closed Brayton cycle	50					1/5	50	Skip if

Section A: Fundamentals, load curves, economics, forecasting, Bangladesh, hydro

questions 1-4 (1-3 in 2020)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A10 Load forecasting: need, factors, methods (similar-day, additive, peak vs energy)

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 79 marks in total

STUDY THESE

- Load forecasting = predicting future demand (MW) and energy (MWh) so plants, fuel and lines are ready in time
- Factors: time (hour, day, season, holiday), weather (temperature, humidity), economy and population, price, special events, random
- Short-term (hours-week): similar-day, regression, time series, ANN; medium (weeks-year); long-term (years): peak-load approach vs energy approach; additive model L = Ln + Lw + Ls + Lr

> 2024 Q3(b) 11 marks
>
> What factors influence load forecasting? Explain the additive model.

> 2024 Q3(c) 5 marks
>
> Why is load forecasting necessary?

> 2023 Q3(a) 10 marks
>
> What is load forecasting? What factors affect it? What are its purposes?

> 2023 Q3(b) 8 marks
>
> What are the methods of short-term load forecasting? Explain the similar-day approach.

> 2022 Q3(c) 10 marks
>
> What is load forecasting? What are the factors affecting it?

> 2022 Q3(d) 8 marks
>
> Describe the importance of load forecasting.

> 2021 Q3(a) 10 marks
>
> What is load forecasting? What are its purposes and the factors affecting it?

> 2021 Q3(b) 6 marks
>
> Distinguish between peak load approach and energy approach of load forecasting.

2020 Q2(b) 11 marks

Why is load forecasting required? Describe its importance.

Full answer: ESE 4115 Study Pack, topic A10.

A4 Effects of variable load on power plant design and operation

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 47 marks in total

STUDY THESE

- Load changes hour to hour, but the plant is cheapest at steady full load
- Design effects: need several units of different sizes, spare (reserve) capacity, quick-start peak plants, flexible boilers/governors
- Operation effects: part-load running lowers efficiency, raises cost per kWh, needs load prediction and good control; total installed capacity > maximum demand

2024 Q1(d) 10 marks

Explain the effect of variable load on a steam power plant with a schematic.

2023 Q2(a) 9 marks

What are the effects of variable load on power plant design and operation?

2022 Q2(a) 9 marks

What are the effects of variable load on power plant design and operation?

2021 Q2(a) 9 marks

What are the effects of variable load on power plant design and operation?

2020 Q1(b) 10 marks

What are the effects of variable load on power plant design and operation?

Full answer: ESE 4115 Study Pack, topic A4.

A2 Load curve, load duration curve and the plant factors (load, demand, diversity, capacity, use)

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 50 marks in total

STUDY THESE

- Load curve = load vs time of day (chronological); load duration curve (LDC) = the same loads sorted from highest to lowest; area under either curve = energy
- Load factor = average load / maximum demand; demand factor = maximum demand / connected load (≤ 1); diversity factor = sum of individual maximum demands / maximum demand of the whole system (≥ 1)
- Capacity factor = average load / installed capacity; use factor = energy produced / (capacity × hours actually run)

> 2024 Q1(a) 9 marks
>
> Define mathematically load factor, diversity factor and demand factor. Which of them are more than 1 and which less than 1?

> 2024 Q1(b) 6 marks
>
> Distinguish between plant capacity factor and plant use factor.

> 2023 Q1(a) 6 marks
>
> Distinguish between ideal load curve and realized load curve.

> 2022 Q1(b) 6 marks
>
> Define: (i) plant use factor, (ii) demand factor.

> 2022 Q1(c) 6 marks
>
> Distinguish between load curve and load duration curve.

> 2021 Q1(b) 8 marks
>
> Describe a typical daily load curve of Bangladesh.

> 2021 Q1(c) 9 marks
>
> Define: (i) plant capacity factor, (ii) diversity factor, (iii) demand factor.

Full answer: ESE 4115 Study Pack, topic A2.

A9 Tariff: objectives, requirements and types (flat, block, two-part, maximum demand, ToU)

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 37 marks in total

STUDY THESE

- Tariff = the rule by which the electricity bill is calculated; must recover fixed + running cost + profit

- General form Y = DX + EZ + C: D per kW of maximum demand, E per kWh, C fixed per customer
- Straight line meter rate = one price per kWh for all units; step meter rate = whole consumption billed at the rate of the step it falls in; block rate = first block one price, next block another

> 2024 Q3(a) 9 marks
>
> What are the objectives and requirements of a tariff?

> 2023 Q2(c) 8 marks
>
> What is meant by meter rate? Compare straight line meter rate and step meter rate tariffs.

> 2022 Q2(d) 8 marks
>
> What is tariff? What are its different types?

> 2021 Q3(c) 12 marks
>
> What is meant by tariff? Write the objectives and requirements of a tariff.

Full answer: ESE 4115 Study Pack, topic A9.

A7 Cost of generation, fixed and operating costs, depreciation

Must study · Very likely · 87% · Easy

Asked in 4 of 5 papers · 38 marks in total

STUDY THESE

- Fixed cost (paid even if no power is made): interest on capital, depreciation, insurance, taxes, fixed staff salaries
- Operating (running) cost: fuel, lubricant, water, wages tied to output, maintenance and repairs
- Depreciation = loss of value with time; methods: straight line, sinking fund, diminishing value. Factors: wear and tear, ageing, obsolescence, inadequacy, accidents

> 2024 Q2(b) 10 marks
>
> Explain fixed cost and operating cost of a power plant.

> 2023 Q2(b) 8 marks
>
> What is meant by depreciation of a power plant? What factors influence depreciation?

> 2022 Q2(c) 12 marks
>
> Describe the costs involved in the generation of electrical energy.

> 2020 Q2(a) 8 marks
>
> What is meant by depreciation? What factors influence depreciation?

Full answer: ESE 4115 Study Pack, topic A7.

A5 Site selection for a power plant and the interconnected grid

Must study · Very likely · 80% · Easy

Asked in 4 of 5 papers · 46 marks in total

STUDY THESE

- Site factors: fuel supply, water, land (cheap, strong ground), transport, distance to load, ash disposal, environment, labour, future expansion
- Interconnected grid = many plants tied together by transmission lines so they share load
- Grid benefits: less reserve needed, uses the cheapest plants first, reliability, diversity of loads, big efficient units, exchange of surplus power

> 2024 Q1(c) 10 marks
>
> What factors are considered when selecting the site of a power plant?

> 2023 Q1(d) 10 marks
>
> What are the advantages of an interconnected grid system?

> 2021 Q2(b) 8 marks
>
> What factors should be considered while selecting the site for a power plant?

> 2021 Q2(c) 10 marks
>
> Describe the benefits of an interconnected grid system.

> 2020 Q1(c) 8 marks
>
> What factors should be considered while selecting the site for a power plant?

Full answer: ESE 4115 Study Pack, topic A5.

A12 Hydro-electric plant: layout, essential features, dam,

Must study · Very likely · 100% · Medium

spillway, surge tank, forebay, hydrology terms, classification

Asked in 5 of 5 papers · 73 marks in total

STUDY THESE

- Water stored high behind a dam falls through a penstock, spins a turbine, drives a generator, leaves by the tail race. P = ρgQHη
- Parts: catchment area, reservoir, dam (+ spillway), intake/gate, pressure tunnel, surge tank, forebay, penstock, power house, draft tube, tail race
- Hydrograph = discharge vs time; flow duration curve = discharge vs % of time; mass curve = cumulative volume vs time; run-off = rain that reaches the stream

> `2024 Q4(a) 9 marks`
>
> Write the essential features of a hydro plant. What are the functions of surge tank and forebay?

> `2024 Q4(b) 8 marks`
>
> What is the function of a dam? Describe the various types of dam.

> `2023 Q4(a) 9 marks`
>
> Define: (i) surge tank, (ii) spillway, (iii) hydrograph.

> `2023 Q4(b) 8 marks`
>
> Classify hydro-electric power plants.

> `2022 Q4(a) 10 marks`
>
> Describe the working principle of a hydro-electric plant with a neat sketch. Write the functions of surge tank and forebay.

> `2022 Q4(b) 9 marks`
>
> Define: (i) run-off, (ii) hydrograph, (iii) specific speed.

> `2021 Q4(a) 10 marks`
>
> Describe the essential features of a hydro-electric power plant with a neat sketch.

> `2021 Q4(c) 4 marks`
>
> Define hydrograph and mass curve.

> `2020 Q3(b) 6 marks`
>
> Why is hydropower called a renewable source of energy?

Full answer: ESE 4115 Study Pack, topic A12.

A14 Numericals: hydro power from catchment, rainfall and head

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 62 marks in total

STUDY THESE

- Water per year (m^3) = catchment area (m^2) × rainfall (m) × fraction that reaches the plant
- Flow Q (m^3/s) = yearly volume / (365 × 24 × 3600 s)
- Power P (kW) = 9.81 × Q × H × η (ρ = 1000 kg/m^3 is already inside 9.81); plant capacity = average power / load factor

> 2024 Q4(d) 10 marks
>
> Reservoir 250 m above the turbine, annual replenishment 70×10^{11} kg, head lost 30 m, overall efficiency 85 %. Find the energy available at the bus-bars per year.

> 2023 Q4(d) 11 marks
>
> Catchment 800 km^2, rainfall 185 cm/year, head 35 m, infiltration + evaporation loss 42 %, turbine efficiency 84 %, generator efficiency 85 %, load factor 0.5. Find power developed and suggest turbines.

> 2022 Q4(d) 11 marks
>
> Rainfall 800 mm, catchment 150 km^2, head 200 m, yield factor 60 %, overall efficiency 75 %, annual load factor 45 %. Find (i) average power, (ii) plant capacity.

> 2021 Q4(d) 15 marks
>
> Catchment 14×10^8 m^2, rainfall 125 cm/year, 70 % available, head 40 m, 5 % head lost in penstock, turbine efficiency 85 %, generator 90 %, annual load factor 45 %. Find the maximum plant capacity.

> 2020 Q3(d) 15 marks
>
> Catchment 800 km^2, rainfall 180 cm/year, head 32 m, loss 40 %, turbine efficiency 85 %, generator 90 %, load factor 0.5. Find power developed and suggest turbines.

Full answer: ESE 4115 Study Pack, topic A14.

A3 Numericals on load curve, load factor, capacity factor, use factor

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 55 marks in total

STUDY THESE

- Energy (MWh) = Σ load × hours = area under the load curve; average load = energy / hours
- Load factor = average / maximum demand; capacity factor = average / installed capacity; reserve = installed − maximum demand
- Use factor = energy / (capacity × hours in service), so hours in service = 8760 × capacity factor / use factor

> 2024 Q2(c) 15 marks
>
> The annual load duration curve of a station is a straight line from 20 MW to 4 MW. Installed: 2 × 10 MW + 1 × 5 MW. Find installed capacity, plant factor, units generated per year, load factor and utilisation factor.

> 2023 Q1(e) 9 marks
>
> Maximum demand 25 MW, load factor 60 %, plant capacity factor 50 %, plant use factor 74 %. Find (i) reserve capacity, (ii) daily energy produced, (iii) maximum energy that could be produced daily if the plant ran fully loaded as per its schedule.

> 2022 Q1(d) 11 marks
>
> Daily load cycle: 300 MW for 8 h, 150 MW for 6 h, 130 MW for 3 h, 100 MW for 7 h. Station has 2 sets of 200 MW. Find (i) daily load factor, (ii) plant capacity factor, (iii) daily fuel needed if CV = 10,000 kcal/kg and heat rate = 25,000 kcal/kWh.

> 2021 Q1(d) 12 marks
>
> Daily load: 6-8 h 15 MW, 8-12 h 35 MW, 12-16 h 65 MW, 16-20 h 25 MW, 20-24 h 60 MW, 24-6 h 20 MW. Draw the load curve and LDC and find energy generated per day.

> 2020 Q1(d) 8 marks
>
> A power station has use factor 45 % and capacity factor 35 %. For how many hours in a year was it not in service?

Full answer: ESE 4115 Study Pack, topic A3.

A13 Hydraulic turbines: selection, specific speed, governing of impulse (Pelton) turbine

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 43 marks in total

STUDY THESE

- Impulse (Pelton) for high head (> 250-400 m); Francis (reaction, radial/mixed) for 50-400 m; Kaplan/propeller (axial) for low head (< 50 m)
- Selection factors: head, specific speed Ns = N√P / $H^{5/4}$, speed (N = 120f/p), part-load efficiency, size, cavitation/setting height, cost
- Governing keeps speed constant when load changes: flyball governor → relay valve → servomotor piston moves the spear (needle) and the deflector

> `2024` `Q4(c)` `8 marks`
> Explain the governing of an impulse turbine.

> `2023` `Q4(c)` `7 marks`
> Which factors should be considered while selecting turbines for a hydro plant?

> `2022` `Q4(c)` `7 marks`
> Which factors should be considered while selecting turbines for a hydro plant? Explain.

> `2021` `Q4(b)` `7 marks`
> Explain the governing of an impulse turbine with a neat sketch.

> `2020` `Q3(a)` `7 marks`
> Which factors should be considered while selecting turbines for a hydro plant?

> `2020` `Q3(c)` `7 marks`
> Why is governing of a hydro-electric power plant required?

Full answer: ESE 4115 Study Pack, topic A13.

A11 Bangladesh power sector: fuel mix, generation cost, renewables, PSMP 2015, nuclear and coal plans

Must study · Very likely · 93% · Medium

Asked in 4 of 5 papers · 62 marks in total

STUDY THESE

- Natural gas is the main fuel (about half of generation), then furnace oil/HFO, coal, imports, diesel, hydro (Kaptai 230 MW), solar

- Cost of generation is lowest for hydro and gas, then coal, highest for HFO and diesel
- PSMP 2015 (Power System Master Plan) projects demand to about 51,000 MW by 2041; met with gas/LNG, imported coal (Rampal, Payra, Matarbari), nuclear (Rooppur 2 × 1200 MW), imports and renewables

> 2024 Q3(d) 10 marks
>
> Describe the future plan for nuclear and coal based generation according to PSMP.

> 2023 Q3(c) 9 marks
>
> Explain the present situation of power generation in Bangladesh from renewable energy sources.

> 2023 Q3(d) 11 marks
>
> Explain the future power demand of Bangladesh according to PSMP 2015. How can the demand be met?

> 2022 Q3(a) 10 marks
>
> Which fuels are widely used for power generation in Bangladesh? Compare the generation cost of BPDB's own plants by fuel type.

> 2022 Q3(b) 11 marks
>
> Explain the future power demand of Bangladesh according to PSMP 2015. How can it be met?

> 2021 Q3(d) 11 marks
>
> Explain the future power demand of Bangladesh according to PSMP 2015. How can it be met?

Full answer: ESE 4115 Study Pack, topic A11.

A1 Power plant basics: classification, energy sources, base vs peak plant, choosing capacity

Must study · Likely · 67% · Easy

Asked in 4 of 5 papers · 53 marks in total

STUDY THESE

- Power plant = a set-up that converts a primary energy source (coal, gas, oil, water, uranium, sun, wind) into electricity; power plant engineering = design, operation, economics of these plants

- Classify by fuel/energy source, by prime mover, by load served (base, intermediate, peak), by location (central, captive)
- Base load plant runs all the time at near full load (steam, nuclear, big hydro); peak load plant starts fast and runs a few hours (gas turbine, diesel, pumped storage, small hydro)

> 2023 Q1(b) 9 marks
>
> What is meant by base load power plant and peak load power plant? What types of power plant are suitable for peak load power plant?

> 2023 Q1(c) 6 marks
>
> What factors should be considered while selecting the capacity of a power plant?

> 2022 Q1(a) 12 marks
>
> What is meant by power plant engineering? Describe the classification of power plant. What are the major sources of energy widely used in power generation?

> 2022 Q2(b) 6 marks
>
> What factors should be considered while selecting the capacity of a power plant?

> 2021 Q1(a) 10 marks
>
> What is meant by power plant engineering? Classify power plants. What are the basic sources of energy?

> 2020 Q1(a) 10 marks
>
> What is meant by power plant engineering? Describe the classification of power plant.

Full answer: ESE 4115 Study Pack, topic A1.

A8 Numericals: generating cost per kWh

Must study · Possible · 47% · Medium

Asked in 3 of 5 papers · 36 marks in total

STUDY THESE

- Energy per year = capacity (kW) × load factor (or capacity factor) × 8760 h
- Yearly cost = (interest + depreciation) % × capital + wages + fuel (Tk/kWh × kWh)
- Cost per kWh = total yearly cost / kWh sent out (subtract auxiliary use if given)

> 2023 Q2(d) 11 marks

> Plant 100 MW, annual load factor 45 %, capital cost Tk 105 crore, annual wages Tk 40 lac, fuel + maintenance Tk 1/kWh, interest 4 %, depreciation 6 %. Find generating cost per kWh.

> 2021 Q2(d) 12 marks
>
> Installed 100 MW, capital cost Tk 25,000/kW, interest + depreciation 10 %, annual load factor 50 %, annual capacity factor 45 %, annual running charge Tk 500 × 10^6, auxiliaries 5 %. Find generating cost per kWh.

> 2020 Q2(c) 13 marks
>
> Plant 100 MW, annual load factor 45 %, capital cost Tk 150 crore, annual wages Tk 1 crore, fuel Tk 0.5/kWh, interest 6 %, depreciation 8 %. Find generating cost per kWh.

Full answer: ESE 4115 Study Pack, topic A8.

A6 Economic load sharing between a base load and a peak load plant (derivation)

Skip if short · Possible · 33% · Hard

Asked in 1 of 5 papers · 10 marks in total

STUDY THESE

- Each plant's yearly cost = fixed part (per kW installed) + running part (per kWh)
- Base plant: high fixed, low running cost; peak plant: low fixed, high running cost
- Peak plant should carry the loads that last less than t = (P1 − P2)/(R2 − R1) hours a year (P = fixed Tk/kW, R = running Tk/kWh)

> 2024 Q2(a) 10 marks
>
> Derive the condition for economic load sharing between a base load plant and a peak load plant.

Full answer: ESE 4115 Study Pack, topic A6.

Section B: Diesel, gas turbine, combined cycle, steam plant and auxiliaries

questions 5-8 (4-6 in 2020)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B1 Diesel power plant: layout, working, fuel system, applications, advantages, peak-load use, lubrication

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 97 marks in total

STUDY THESE

- Diesel engine (compression ignition) drives an alternator; 2-50 MW; used for standby, peak, mobile, small isolated loads
- Systems: fuel supply (tank → day tank → filter → injection pump → injector), air intake + exhaust, cooling, lubrication, starting, governing
- Solid injection = fuel is pressurised by a pump and sprayed directly (no air compressor) → simpler, cheaper than air-blast injection

> 2024 Q5(a) 5 marks
>
> What are the advantages of a diesel power plant as a peak load plant?

> 2024 Q5(b) 5 marks
>
> Why is the solid injection fuel system more widely used in diesel engines?

> 2023 Q5(a) 12 marks
>
> With a schematic, describe the working principle of a diesel engine power plant.

> 2023 Q5(b) 10 marks
>
> What are the applications of diesel power plants? Explain.

> 2022 Q5(a) 11 marks
>
> Draw the typical layout of a diesel engine power plant. Explain its fuel supply system.

> 2022 Q5(b) 9 marks
>
> Write the advantages and disadvantages of a diesel power plant.

> 2021 Q5(a) 20 marks
>
> With a neat sketch, describe the working principle of a diesel engine power plant.

> 2021 Q5(b) 6 marks
>
> Write the importance of the lubrication system in a diesel power plant.

> 2021 Q5(c) 9 marks

> Explain the fuel supply system of a diesel plant.

> 2020 Q4(a) 10 marks
> Is a diesel power plant suitable as a peak load plant? Describe the working of a diesel plant.

Full answer: ESE 4115 Study Pack, topic B1.

B10 Feed water: impurities, troubles in boiler, treatment methods (lime-soda, zeolite)

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 56 marks in total

STUDY THESE

- Impurities: dissolved salts (Ca, Mg → hardness), dissolved gases (O_2, CO_2), suspended matter, acids, oil
- Troubles: scale (overheating, tube burst), corrosion, carry-over/priming/foaming, caustic embrittlement
- Treatment: mechanical (sedimentation, coagulation, filtration), thermal (deaeration, evaporation), chemical (lime-soda, zeolite/ion exchange), demineralisation, blow-down

> 2024 Q8(a) 6 marks
> Discuss the adverse effects of using raw water in a boiler.

> 2024 Q8(b) 12 marks
> What methods are used to treat water in a power plant? Explain one commonly used method.

> 2023 Q8(a) 11 marks
> Why is treatment of raw water necessary? Explain the zeolite process of softening feed water.

> 2022 Q8(a) 12 marks
> What impurities may be present in natural water? What problems do they cause in a power plant?

> 2021 Q8(a) 15 marks
> What are the methods of feed water treatment? Describe the chemical treatment method with a sketch.

Full answer: ESE 4115 Study Pack, topic B10.

B6 Combined cycle power plant (gas turbine + steam turbine)

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 49 marks in total

STUDY THESE

- Gas turbine exhaust (~500-600 °C) goes to a heat recovery steam generator (HRSG) that makes steam for a steam turbine
- Both turbines make power from the same fuel → efficiency 50-62 % vs 30-40 % for either alone
- T-s: Brayton cycle on top, Rankine cycle below, linked by the HRSG heat transfer

> 2024 Q7(c) 10 marks
>
> What are the advantages of a combined cycle power plant? Draw the P-v and T-s diagrams.

> 2023 Q6(c) 10 marks
>
> What is meant by a combined cycle power plant? What are its advantages?

> 2022 Q6(c) 14 marks
>
> Explain the combined working of gas turbine and steam power plant.

> 2021 Q6(a) 15 marks
>
> What is a combined cycle power plant? Explain with a diagram.

Full answer: ESE 4115 Study Pack, topic B6.

B4 Improving gas turbine efficiency: regeneration, intercooling, reheating (and all three together)

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 65 marks in total

STUDY THESE

- Regeneration: hot turbine exhaust preheats compressed air → less fuel for the same work → efficiency up (work unchanged)

- Intercooling: cool air between compressor stages → less compressor work → net work up; reheating: reheat gas between turbine stages → more turbine work
- Intercooling and reheating alone raise work but lower efficiency; with a regenerator they raise both

> 2024 Q6(b) 10 marks
> With P-v and T-s diagrams, analyse the performance of a gas turbine plant when an intercooler, a reheater and a regenerator are used together.

> 2023 Q7(a) 12 marks
> Explain the effect of regeneration on the work output and efficiency of a gas turbine plant.

> 2022 Q6(b) 16 marks
> Explain the methods used to improve the thermal efficiency of a gas turbine plant. Is it possible to increase efficiency using exhaust gas?

> 2021 Q7(a) 15 marks
> What are the purposes of intercooling and reheating in a gas turbine plant? Describe.

> 2020 Q5(a) 12 marks
> Explain the various methods used for improvement of thermal efficiency of a gas turbine power plant.

Full answer: ESE 4115 Study Pack, topic B4.

B11 Draught: natural vs mechanical (forced, induced, balanced), chimney and its height formula

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 39 marks in total

STUDY THESE

- Draught = small pressure difference that pushes air through the fuel bed and drives flue gas out of the chimney
- Natural draught from chimney: hot light gas inside, cold heavy air outside; $\Delta p = H g (\rho_{air} - \rho_{gas})$
- Mechanical: forced (FD fan at inlet), induced (ID fan before chimney), balanced (both, furnace at slightly below atmospheric)

> 2024 Q8(d) 5 marks
>
> Why are draught and chimney used in a power plant?

> 2023 Q8(b) 8 marks
>
> What is draught? Compare natural and mechanical draught.

> 2022 Q8(b) 8 marks
>
> What is a balanced draught system? Describe its working with a neat sketch.

> 2021 Q8(b) 10 marks
>
> Write the purpose of a chimney. Derive an expression for the height of a chimney.

> 2020 Q6(a) 8 marks
>
> What is draught? Write its importance. Compare natural and mechanical draught.

Full answer: ESE 4115 Study Pack, topic B11.

B13 Cooling towers: types, induced-draught counterflow wet tower, range, approach, efficiency + numerical

Must study · Very likely · 93% · Medium

Asked in 4 of 5 papers · 46 marks in total

STUDY THESE

- Hot condenser water is sprayed down over fill; air moves up; a little water evaporates and the rest cools
- Range = water in − water out; approach = water out − wet bulb temperature of air in; efficiency = range / (range + approach)
- Types: natural draught (hyperbolic), mechanical: forced draught, induced draught (counterflow or crossflow); wet vs dry

> 2024 Q8(c) 12 marks
>
> Explain the general working principle of a wet cooling tower.

> 2023 Q8(c) 9 marks
>
> Write short notes on (i) approach, (ii) range, (iii) cooling efficiency of a cooling tower.

> 2022 Q8(c) 15 marks
>
> What is a cooling tower? Classify it. Describe the induced draught counterflow wet cooling tower with a schematic.

> 2021 Q8(c) 10 marks
>
> Water at 30 °C enters a cooling tower at 1.25 kg per kg of air. Air enters at 20 °C, 60 % RH and leaves at 28 °C, 90 % RH. Make-up water is at 20 °C. Find (i) water outlet temperature, (ii) fraction of water evaporated, (iii) approach and range.

Full answer: ESE 4115 Study Pack, topic B13.

B3 Gas turbine plant: open-cycle layout, open vs closed, applications, peak-load advantages, fuel properties

Good to have · Likely · 60% · Easy

Asked in 3 of 5 papers · 25 marks in total

STUDY THESE

- Open cycle: air → compressor → combustion chamber (fuel burnt) → turbine → exhaust to atmosphere; turbine drives compressor + generator
- Closed cycle: same working fluid circulates; heat added in a heater and removed in a cooler
- Peak-load advantages: starts in minutes, small, cheap, little water, no standby loss

> 2023 Q6(a) 7 marks
>
> What are the advantages of a gas turbine plant as a peak load plant?

> 2023 Q6(b) 8 marks
>
> Draw the layout of an open cycle gas turbine power plant.

> 2022 Q6(a) 5 marks
>
> Write the applications of gas turbine power plants.

> 2021 Q6(c) 5 marks
>
> What properties are considered while selecting fuel for a gas turbine plant?

Full answer: ESE 4115 Study Pack, topic B3.

B8 Steam power plant: layout and four circuits, working, base-load advantages, site, design activities

Must study · Likely · 67% · Medium

Asked in 4 of 5 papers · 76 marks in total

STUDY THESE

- Rankine cycle: pump → boiler → turbine → condenser → pump
- Four circuits: coal and ash; air and flue gas; feed water and steam; cooling water
- Good base-load plant: large unit sizes, cheap fuel (coal), long life, steady output at best efficiency

> 2023 Q7(b) 15 marks
>
> With a neat sketch describe the working principle of a steam power plant.

> 2023 Q7(c) 8 marks
>
> What are the advantages of a steam power plant as a base load plant?

> 2022 Q7(a) 15 marks
>
> Draw a general layout of a steam power plant and discuss its various circuits.

> 2021 Q7(b) 15 marks
>
> Draw the typical layout of a steam power plant and explain its working.

> 2021 Q7(c) 10 marks
>
> What activities should be carried out for designing a steam power plant?

> 2020 Q6(b) 13 marks
>
> Describe the factors to be considered while selecting the site for a steam power plant.

Full answer: ESE 4115 Study Pack, topic B8.

B5 Brayton cycle numericals and the maximum-work condition

Must study · Likely · 67% · Hard

Asked in 4 of 5 papers · 51 marks in total

STUDY THESE

- $T2/T1 = T3/T4 = rp^{(\gamma-1)/\gamma} = rp^{0.286}$ for air (isentropic); actual: $T2 = T1 + (T2s - T1)/\eta c$, $T4 = T3 - \eta t(T3 - T4s)$
- $wc = cp(T2 - T1)$, $wt = cp(T3 - T4)$, $wnet = wt - wc$, $q_in = cp(T3 - T2)$, $\eta = wnet/q_in$, back work ratio $= wc/wt$
- Maximum net work when $T2 = T4 = \sqrt{(T1T3)}$; ideal $\eta = 1 - 1/rp^{0.286}$

> 2023 Q6(d) 10 marks

> Show that the net work output of a gas turbine plant is maximum when the exit temperatures of compressor and turbine are the same.

> 2022 Q7(c) 12 marks
>
> Ideal gas turbine with two stages of compression and two of expansion, pressure ratio 2.83 per stage, air enters each compressor stage at 300 K and each turbine stage at 1300 K. Find back work ratio and thermal efficiency (i) without regenerator, (ii) with an ideal regenerator.

> 2021 Q6(b) 15 marks
>
> Brayton cycle: pressure ratio 8, compressor inlet 300 K, turbine inlet 1300 K, ηc = 80 %, ηt = 85 %. Find (i) exit temperatures, (ii) back work ratio, (iii) thermal efficiency, (iv) efficiency with an 80 % effective regenerator.

> 2020 Q5(b) 14 marks
>
> Ideal Brayton cycle: pressure ratio 9, compressor inlet 310 K, turbine inlet 1100 K. Using air-standard assumptions find the thermal efficiency.

Full answer: ESE 4115 Study Pack, topic B5.

B2 Diesel plant cooling (necessity, thermostat cooling) and starting methods

Must study · Possible · 53% · Easy

Asked in 3 of 5 papers · 53 marks in total

STUDY THESE

- Gas in the cylinder reaches ~2750 °C; 25-35 % of fuel heat must be removed or oil film burns, metal weakens, pistons seize
- Thermostat cooling: a thermostat valve keeps water in the jacket until it is hot (~70-80 °C), then opens to the radiator; prevents over-cooling
- Starting: compressed air (big engines), electric motor + battery (small), auxiliary petrol engine

> 2023 Q5(c) 10 marks
>
> What are the methods used to start a diesel power plant? Briefly explain.

> 2023 Q5(d) 8 marks
>
> What are the necessities of a cooling system in a diesel plant? Describe thermostat cooling with a figure.

> 2022 Q5(c) 15 marks

> What is the necessity of a cooling system in a diesel plant? Describe the thermostat cooling system with a figure.

> 2020 Q4(b) 10 marks
>
> Write the disadvantages of a diesel power plant. Why is cooling necessary?

> 2020 Q4(c) 10 marks
>
> Describe the methods used to start a diesel power plant.

Full answer: ESE 4115 Study Pack, topic B2.

B9 Reheat Rankine cycle and binary vapour cycle

Good to have · Possible · 53% · Medium

Asked in 2 of 5 papers · 36 marks in total

STUDY THESE

- Reheat: steam expands in HP turbine, returns to boiler to be reheated, then expands in LP turbine → drier exhaust, more work, slightly higher η
- Binary vapour cycle: mercury cycle on top (high temperature at low pressure), its condenser boils water for a steam cycle below
- Not widely used: mercury is costly, toxic, scarce, does not wet metal; complex plant; modern supercritical steam and combined cycles do better

> 2024 Q7(a) 15 marks
>
> Explain the working principle of a steam power plant on the reheat Rankine cycle with a schematic.

> 2024 Q7(b) 10 marks
>
> Why is a binary vapour cycle used in a power plant? Why is it not widely used?

> 2022 Q7(b) 11 marks
>
> With a schematic describe the working of a binary vapour cycle power plant.

Full answer: ESE 4115 Study Pack, topic B9.

B12 Numericals: chimney draught and chimney height

Good to have · Possible · 33% · Medium

Asked in 2 of 5 papers · 17 marks in total

STUDY THESE

- Draught in mm of water: h_w = 353 H [1/T_a − ((m+1)/m)(1/T_g)] (m = kg air per kg fuel, T in K)
- Height of hot gas column: H_g = H[(m/(m+1))(T_g/T_a) − 1]
- Using densities: ρ = p/(RT); draught (kgf/m^2 = mm of water) = H (ρ_air − ρ_gas)

> 2023 Q8(d) 8 marks
>
> Chimney 28 m high, hot gas 320 °C, outside air 27 °C, 15 kg air per kg coal. Find (i) draught in mm of water, (ii) draught as height of hot gas column.

> 2020 Q6(c) 9 marks
>
> Find chimney height to produce a static draught of 22 mm of water. Mean flue gas temperature 290 °C, ambient 20 °C, R_air = 29.26 and R_gas = 26.2 kgf·m/kg·K, barometer 760 mm Hg.

Full answer: ESE 4115 Study Pack, topic B12.

B7 Exergy analysis numericals (2024): Diesel cycle and closed Brayton cycle

Skip if short · Possible · 33% · Hard

Asked in 1 of 5 papers · 50 marks in total

STUDY THESE

- Exergy = the useful part of energy relative to the surroundings (dead state T0, P0); destroyed exergy = T0 × entropy generated
- For each process: energy balance (q − w = Δu or Δh) and entropy balance (s_gen = Δs − q/T_boundary)
- Energy efficiency = w_net / q_in; exergy (second-law) efficiency = w_net / [q_in(1 − T0/T_source)]

> 2024 Q5(c) 25 marks
>
> A diesel plant operates on a simple Diesel cycle with air, compression ratio 16. Air is at 323 K and 100 kPa at the start of compression and 1723 K at the end of heat addition. Compression efficiency 85 %, expansion efficiency 90 %, T_source = 2000 K, T0 = 300 K. (i) Draw schematic and T-s; (ii) write mass, energy, entropy and exergy balances for each process; (iii) find all state properties (P, T, v, u, h, s, e); (iv) total entropy generation, exergy destruction, net work, energy and exergy efficiency.

> 2024 Q6(a) 25 marks

> A closed gas turbine plant on the simple Brayton cycle has pressure ratio 16. Air enters the compressor at atmospheric pressure and 300 K at 10 kg/s; it leaves the turbine at 973 K; the gas exits the combustion chamber at 1800 K. Compressor and turbine isentropic efficiency 85 %, variable specific heats, T_s = 2500 K, T0 = 293 K, P0 = 100 kPa. (i) Schematic and T-s; (ii) balances for all components; (iii) state properties; (iv) total entropy generation, exergy destruction, net work, energy and exergy efficiency.

Full answer: ESE 4115 Study Pack, topic B7.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AC	alternating current
ANN	artificial neural network
ARIMA	auto-regressive integrated moving average
ARMA	auto-regressive moving average
BHP	brake horsepower
BIFPCL	Bangladesh-India Friendship Power Company Ltd
BPDB	Bangladesh Power Development Board
CCGT	combined cycle gas turbine
CCPP	combined cycle power plant
CF	capacity factor
CHP	combined heat and power (cogeneration)
CI	compression ignition (diesel)
CPGCBL	Coal Power Generation Company Bangladesh Ltd
CV	calorific value (heat released by 1 kg of fuel)
DBT	dry bulb temperature
EGCB	Electricity Generation Company of Bangladesh
ESP	electrostatic precipitator

FBC	fluidised bed combustion
FD	forced draught
GT	gas turbine
GWh	gigawatt-hour = 10^6 kWh
HCV	higher calorific value
HFO	heavy fuel oil
HP	high pressure
HRSG	heat recovery steam generator
HSD	high speed diesel
ID	induced draught
IP	intermediate pressure
IPP	independent power producer
kW	kilowatt = 1000 W
kWh	kilowatt-hour = 1 unit of electrical energy
LDC	load duration curve
LF	load factor
LNG	liquefied natural gas
LP	low pressure
LTLF	long-term load forecasting
MTLF	medium-term load forecasting
MW	megawatt = 1000 kW
MWh	megawatt-hour = 1000 kWh
NOx	nitrogen oxides
NWPGCL	North-West Power Generation Company Ltd
PSMP	Power System Master Plan
PV	photovoltaic (solar cells)
RE	renewable energy
RH	relative humidity
RPCL	Rural Power Company Ltd

rpm	revolutions per minute
SI	spark ignition (petrol)
SOx	sulphur oxides
ST	steam turbine
STLF	short-term load forecasting
TDC	top dead centre
Tk	Taka (Bangladeshi currency)
ToU	time of use
TWL	tail water level
UF	use (utilisation) factor
WBT	wet bulb temperature

Built from the scanned ESE 4115 papers (2020, 2021, 2022, 2023, 2024). Question text is cleaned up from the scans.